CÉLESTIN **DEMUYTERE**

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Professional Experience

Ghent University (UGhent) May 2022 – current

PhD Researcher STEN – Faculty Bioscience Engineering

- Doctoral research on the circularity, environmental impact and criticality of offshore wind decommissioning (promotor Prof. Jo Dewulf)
- **DECOMPASS** (Energy Transition Fund, 2025-2028): UGent lead on foundation decommissioning techniques (WP2) and LCA and circularity (WP4)
- **ORESA** (Interreg North-West Europe): coordination of the LCA and circularity learning hub and SME support in the offshore renewable energy accelerator
- **CTO** (Flemish ICON project, completed): Circular Transition of Offshore Windmill Materials
- Managing research on MFA, circularity indices, eco-design, critical materials and LCSA
- Develop EoL scenarios for OWF decommissioning as well as LCSA on offshore wind energy
- Supervision of master's theses (ranging from offshore wind cable circularity to critical raw material demand for EU electrification infrastructure)

Vrije Universiteit Brussel (VUB) Jan 2026 – current

Joint PhD Researcher – BUTO, Sustainable Systems Management

- Joint doctorate with UGent as main institution (co-promotor Prof. Gwenny Thomassen)
- Faculty of Social Sciences and Solvay Business School

DFSE (Drone, Freelance & Sustainability engineering – S/E) July 2021 – July 2024

- Drone-oriented services
- Freelance consulting on sustainable development, resource transition and assessment studies

Organic waste systems (OWS) NV Jul 2020 – May 2022

Project Manager/Engineer Sustainability Assessment Services

- Managing LCSA projects on European, Flemish and commercial level
- Bridging knowledge between impact studies, circularity metrics and sustainable development
- Responsible for peer-review studies as well as revising circularity/specific biobased metrics
- Consulting on compostables, bioplastics, microplastics and developments in emerging markets

Artemis milieu Jan 2017 – Jul 2020

Project leader – Soil/groundwater remediation research and treatment

- Leading projects and reporting on soil and groundwater contaminations
- On-site coordination of complex remediation processes
- Modeling contaminations, treatments and toxicology by use of specialized software

PaInT Department (UGent) 2014

Internship – Bioscience Engineering department laboratory

- Laboratory experience concerning Reverse Osmosis and Wastewater treatment
- Designed and executed in context of scientific article publication

Education

Ghent University & Vrije Universiteit Brussel May 2022 – 2026 (exp.)

Joint PhD: Doctor of Bioscience Engineering (UGent) and Doctor of Interdisciplinary Studies: Economics and Environmental Sciences (VUB)

- **Dissertation:** From seabed to smelter: Circularity, environmental impact and material criticality in offshore wind decommissioning

Ghent University Sept 2011 – Jul 2016

Master of Bioscience Engineering: Environmental Technology

- **Thesis:** Acoustics for a Sustainable Future (Prof. Dr. ir. Kathy Steppe, Laboratory of Plant Ecology)
- **Bachelor thesis:** Reconstruction of water-use efficiency, influence of climate change

Publications

- Vetters, J., **Demuytere, C.**, Louis, M., Van Passel, S., Thomassen, G. (2026). Split incentives in shared waters: Introducing a multi-actor techno-economic assessment framework for offshore wind decommissioning. Applied Energy, 418, 128065.
- Godoy León, M.F., Balsells Llort, C., **Demuytere, C.**, Bobba, S., Tazi, N., Dewulf, J. (2025). Multilayer Material System Analysis of wind turbines: correlation of stocks and flows in the EU of six metals and two drivetrain technologies. Journal of Cleaner Production, 518, 145794.
- **Demuytere, C.**, Thomassen, G., Dewulf, J. (2025). Navigating towards environmental sustainability assessments of offshore wind farms at the end-of-life. Waste Management, 205, 115016.
- **Demuytere, C.**, Vanderveken, I., Thomassen, G., Godoy León, M.F., De Luca Peña, L.V., Blommaert, C., Vermeir, J., Dewulf, J. (2024). Prospective material flow analysis of the end-of-life decommissioning: Case study of a North Sea offshore wind farm. Resources, Conservation and Recycling, 200, 107283.
- Vadoudi, K., Deckers, P., **Demuytere, C.**, Askanian, H., Verney, V. (2022). Comparing a material circularity indicator to life cycle assessment: The case of a three-layer plastic packaging. Sustainable Production and Consumption, 33, 820-830.
- Vanoppen, M., Stoffels, G., **Demuytere, C.**, Bleyaert, W., Verliefde, A. (2015). Increasing RO efficiency by chemical-free ion-exchange and Donnan dialysis: Principles and practical implications. Water Research, 80, 59-70.
- Vetters, J., **Demuytere, C.**, Van Passel, S., Thomassen, G. (2026). From Water to Resource: A Multi-Actor Techno-Economic Assessment for the Offshore Wind End-of-Life Supply Chain. SSRN preprint, doi:10.2139/ssrn.7319858.
- **Demuytere, C.**, Vetters, J., Blommaert, C., Vermeir, J., Dewulf, J., Thomassen, G. (under revision). Setting Sail for Sustainable Offshore Wind: Evaluating Material Circularity and Environmental Impacts Across End-of-Life Scenarios for a North Sea Offshore Wind Farm. Waste Management.

Extracurricular

• **Certified courses:** Simapro & LCA (Pré Sustainability), Drone pilot class 1, Asbestos & soil remediation

• **Seminars:** Paradigm shift towards resource recovery, Sustainable Development, Managing an organization & dealing with employees and volunteers (2014)

• **Involved organisations:** UGent1010 (BE 2013-2015, Management committee), StuJardin (BE 2013-2015, Management committee + Director), Bergwaldprojekt (CH 2014, Biodiversity project volunteer)

Skills & Interests

Languages:	Dutch (native), English and German (fluent), French (functional)
Software:	SimaPro, OpenLCA, AutoCAD, S-Risk, Python, R
Methods:	MFA, LCA, circularity indicators, criticality assessment (CriticS)
Interests:	Sustainable transition on individual level, AI integration and public engagement